

Scope or Specifications

Convert Functional Specifications into a Prototype of the system. The system is described in a static form, showing objects' attributes, functions and relationships.

- ✓ Create Data Structures, ERD and Documents
- ✓ Create User Interfaces Diagrams and Documents
- ✓ Create Classes and/or Objects UML Diagrams and Documents
- ✓ Create Components Diagram and Documents
- ✓ Create Deployment Diagrams and Documents
- ✓ Create a GLOSSARY

- Ensure that the specifications are consistent with the system requirements, by setting up a requirements traceability document. This document lists the various requirements, and then tracks how they are addressed in the written specs, the system design (which data flow process addresses the requirement), and the code (which function or subroutine satisfies the requirement).

- Ensure that specifications have been supplied for system flexibility, maintainability, and performance where appropriate.

- Ensure that a testing strategy has been established.

- Ensure that a realistic development schedule has been established, including scheduled reviews.

- Ensure that a formal change procedure has been devised for the system. Uncontrolled changes, resulting in many sequential (or even concurrent) versions of a system, can really contribute to quality degradation.

Planning Phase

The planning phase culminates in the vision/scope approved milestone. At the end of this phase, the project team and the customer agree on the overall direction for the project, including what the product will and will not include. Determining vision and scope are opposite activities, yet a successful project requires both. Vision is the unbounded view of what the solution could be, without regard to project constraints. Scope defines what parts of the vision can be accomplished within the project constraints. A shared project vision is essential for project success.

Deliverables

- Functional Specification
- Functional Specification Signoff
- Risk Assessment

Functional Specification



Owners and List of Contacts

Name	Email	Phone	Role
John Doe	Jdoe@Me.com	303-894-7315 W 303-471-8344 H 303-203-5567 Pgr	Project Manager Development Lead
Joe Tester			System Test Lead
Jane ProdSupport			Production Support Mgr
Joe UserMgr			User Test Lead
Joe Developer			Developer – Presentation Tier
Jane Developer			Developer – Business Tier
Joe DBA			Data Base Administrator
Joe Tester			Tester
Jane Tester			Tester
Joe Customer			Department VP
Jane Customer			Department Mgr
Josey Customer			Product Support

Signoffs

Phase	Name	Date	Signature
Functional Specifications	John Doe, PM/DM	xx/xx/xx	
	Joe Tester, System Test Lead		
	Jane ProdSupport, Production Support Mgr		
	Joe User Mgr, UM		
	Joe Customer, Customer		

Revision History

Date	Reason for change(s)	Author(s)
09/15/1988	First Draft	John Doe

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1. Summary

This document describes the features and timeframe desired for [ProductName]...

2. Project Goals, Justification, and Success Criteria

2.1 Project Goals

Explain the goals of the project here...

2.2 Justification

Explain why the project is needed. List revenue impact or cost savings information.

2.3 Success Criteria

List the success criteria for the project. Once the project is complete, it will be measured back to the criteria listed here to determine if the project was successful.

3. Features

3.1 Feature 1

Explain exactly what the feature is to perform in Use Case format. List any constraints, actors, security implications, etc.

3.2 Feature 2, etc.

4. Security Requirements

Explain the security requirements of the system. If needed, assign user groups and define what those groups have access to.

5. Data Conversion Requirements

Explain any requirements in which we must convert data to import into the system. Detail the mapping needed from their system to the new system.

6. Performance and Response Time Requirements

Explain how many concurrent users will be using the system and expected response time.

7. Platform Dependent and Installation Requirements

Explain what platforms the client presentation layer must run on (Windows 95, 98, NT 4, 5, etc) and describe the installation process.

8. Localization Requirements

Explain any requirements that are specific to localization (European date and postal code format, etc.)

9. Parallel Testing Requirements

Explain the areas of the system (if any) that must be run parallel to an existing production system and compared for consistency.

10. Cross System Interface Requirements

Explain the systems this system interact with and detail any requirements for testing with that system to ensure integrity.

11. Data Archival, Backup and Recovery Requirements

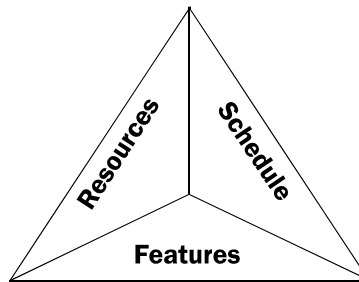
Explain the purge, archival, backup and recovery requirements of the system.

12. Reporting Requirements

Describe the reports (if any) are required for the system. Are they to be online or run in batch and must they be exportable to other formats such as MS Word, MS Excel or HTML.

13. Project Flexibility Matrix

For a project to succeed, there are 3 variables that affect how quickly a project can be completed: **Resources, Schedule and Feature** (see diagram below). One of the sides of the triangle must always be flexible to achieve success. For example, if the client has a fixed number of resources (people that work on the project) and their schedule has been set in stone, the feature set must be flexible. This means that they must be flexible to drop some features to make the pre-determined date with that number of resources.



Project Trade-off Matrix

	Inflexible	Flexible
Resources(Cost)		
Ship Date		
Features		

Working together, the team and customer place a check mark in the appropriate column for each of the project variables. The columns are defined as:

- **Inflexible.** Mark 2 items that are inflexible. For example, if the cost and ship date are set in stone, make Resources (Cost) and Ship Date as Inflexible. Only 2 items can be inflexible, the last item must be flexible.
- **Flexible.** Mark one item as flexible. For example, if you are flexible with the features that are included in the project, mark Features as Flexible.

A team should use the project trade-off matrix as a reference when making decisions. The matrix is not intended to show absolute priorities; it is merely a tool to facilitate communication and understanding. Most important for the project team is that the matrix shows areas in which the customer is willing to compromise. Make sure that no row or column in the project trade-off matrix has more than one check mark. Any other combination poses serious risk to the project and must be accounted for explicitly in the risk management plan.

In order for a team to be successful, at least one check mark must be in the “flexible” column. This means that the team owns one side of the triangle (that iws, owns at least one variable) so that the team is empowered to manage change and risk, and is therefore positioned to achieve success instead of failure.

14. Stack Ranking of Features

To ensure that items are worked on in order of importance, stack rank the features with the lowest stack rank being most important and the highest stack rank being least important. This will allow the development team to focus on the important items and to manage risk.

Ranking	Req #	Feature
1	1.1	Display MSL Listings (very important)
...	...	...
...	...	...
...	...	...
99	1.2	Update agent information (not very important)

15. Roles and Responsibilities

Below are the roles and responsibilities for each phase of the life cycle. Smaller projects may not be broken out to this level of detail.

Life Cycle	Role	Responsibility
Planning	Setup hardware for Development	Development Team
	Functional Specs	Development Team / Client PM
	Detailed Design	Development Team
	Test Design	System Test Lead
	Development Project Plan	Development Team
	Test Project Plan and Budget	System Test Lead
	Overall Project Plan	Project Manager (PM)
Construction	Coding	Development Team
	Unit Testing	Development Team
	System Test - Test Cases	System Test Team
	User Test – User Test Lead	User Test Team
	User Test - Test Cases	User Test Team
	Setup hardware for System Testing	System Test Team
System Testing	Migration of code/database from Development to System Test	Development Team
	Populate test database for System Test	Development Team
	System Testing	System Test Team
	Bug Tracking / Triage	System Test Lead, Development Manager, PM
	Drops for reiteration of fixes	Development Team
User Acceptance Test (UAT)	Migration of code from System Test to UAT	Development Team
	Populate test database for UAT	Development Team
	UAT Testing	User Test Team
	Bug Tracking / Triage	System Test Lead, DM, PM, User Test Lead
	Drops for reiteration of fixes (must go back through System Test)	Development Team
Production	Migration of code from UAT to Production	Development Team

Risk Assessment



As of xx/xx/xx

Risk	Probability of Loss	Size of Loss	Risk Exposure	Contingency Plan
Testers unfamiliar with Application	100%	24h	24h	Built into contingency
Lack of finalized specifications	100%	24h	24h	Need requirements [ProductName] infrastructure changes
Development Slippage of agreed upon dates.	20%	40 hrs	8 hrs	Slippage issues must be addressed in Weekly Status Meetings. No additional time built into schedule for this. If this happens, the testing timeline will have to be updated to reflect the changes.
Quality of development	25%	40 hrs	10 hrs	If the quality of development is low, more iterations of bug fixes must be tracked which could extend the timeline. No additional time built into schedule for this. If this happens, the testing timeline will have to be updated to reflect the changes.

Prior to creating the project plan, we assessed the risks for the testing group. The matrix above lists the risks we identified. We built time into the schedule for many of the risks. Below is an explanation of the matrix:

- **Risk** – This explains what the risk is.
- **Probability of Loss** – This is the probability of the risk actually becoming reality.
- **Size of Loss**– If the risk becomes reality, this is the number of days impact on the project timeline.

- **Risk Exposure** – This is the Probability of Loss * Size of Loss. This gives us an estimated duration in days that we can build into the schedule to hedge our risks.
- **Contingency Plan** – This is the plan of handling the circumstances of the risk.